Problem Set 5

AI 501: Mathematics for Artificial Intelligence | Fall 2026

Problem 1

Let

$$f(x) = \tfrac{1}{2} x^T A x + b^T x,$$

where $A \in \mathbb{R}^{n \times n}$ is symmetric and $b \in \mathbb{R}^n$.

(a) Find the gradient $\nabla f(x)$. *Hint:* differentiate with respect to a single coordinate x_k and use the symmetry of A.

(b) Write the least-squares loss $\mathcal{L}(\theta) = \frac{1}{2}\|y - X\theta\|_2^2$ in the form above (up to a constant), and use part (a) to find its gradient $\nabla \mathcal{L}(\theta)$. *Hint:* expand the squared norm as an inner product.

Solution:

(a) In components, $f(x) = \frac{1}{2}\sum_{i,j} A_{ij} x_i x_j + \sum_i b_i x_i$, so

$$\frac{\partial f}{\partial x_k} = \frac{1}{2}\Big(\sum_j A_{kj} x_j + \sum_i A_{ik} x_i\Big) + b_k = (Ax)_k + b_k,$$

using $A = A^T$. Hence $\nabla f(x) = Ax + b$.

(b) Expanding,

$$\tfrac{1}{2}\|y - X\theta\|_2^2 = \tfrac{1}{2}\theta^T (X^T X)\theta - (X^T y)^T \theta + \tfrac{1}{2} y^T y,$$

which has the form above with $A = X^T X$ (symmetric) and $b = -X^T y$. Therefore

$$\nabla \mathcal{L}(\theta) = X^T X \theta - X^T y.$$

The constant $\frac{1}{2} y^T y$ does not depend on θ, so it does not affect the gradient.

Problem 2

The partial derivatives of $\|Ax\|_2^2$ with respect to $x_1, \ldots, x_n$ form the vector $2A^T Ax$, and the partial derivatives of $2b^T Ax$ form the vector $2A^T b$.

(a) Use these facts to find the condition under which all partial derivatives of $\|Ax - b\|_2^2$ are zero.

(b) Assuming the columns of A are linearly independent, write the minimizer in closed form, and relate it to the linear regression solution $\theta = (X^T X)^{-1} X^T y$.

Solution:

(a) Expanding,

$$\|Ax - b\|_2^2 = x^T A^T A x - 2b^T A x + b^T b,$$

so the gradient is $2A^T Ax - 2A^T b$. Setting it to zero gives the normal equations

$$A^T A \hat{x} = A^T b.$$

(b) If the columns of A are linearly independent, $A^T A$ is invertible and

$$\hat{x} = (A^T A)^{-1} A^T b.$$

With $A = X$, $x = \theta$, $b = y$, this is exactly $\theta = (X^T X)^{-1} X^T y$. The factor $\frac{1}{2}$ in the regression loss only scales the gradient and does not change the minimizer.

Problem 3

We wish to fit a model with only a bias term, $\hat{y} = \theta_0$, to the four observations $y = (0, 8, 8, 20)^T$. An exact fit would require

$$\theta_0 = 0, \qquad \theta_0 = 8, \qquad \theta_0 = 8, \qquad \theta_0 = 20,$$

which is impossible.

(a) Write this system as $X\theta = y$, identifying the 4×1 design matrix X.

(b) Solve the normal equation $X^T X\hat{\theta} = X^T y$ for θ_0.

(c) Find the residual vector $e = y - X\hat{\theta}$ and $\|e\|_2^2$.

(d) What does this tell you about the least-squares prediction of a model with no input features?

Solution:

(a) $X = (1, 1, 1, 1)^T$ and $\theta = \theta_0$.

(b) $X^T X = 4$ and $X^T y = 0 + 8 + 8 + 20 = 36$, so $4\theta_0 = 36$ and $\theta_0 = 9$.

(c) $e = (0 - 9,\ 8 - 9,\ 8 - 9,\ 20 - 9) = (-9, -1, -1, 11)$, and

$$\|e\|_2^2 = 81 + 1 + 1 + 121 = 204.$$

(d) $\theta_0 = 9$ is the sample mean of the y_i. In general, with $X = \mathbf{1}$ the normal equation reads $n\theta_0 = \sum_i y_i$, so a model with only a bias term predicts the mean of the observations.

Problem 4

Fit the line $\hat{y} = \theta_0 + \theta_1 x$ by least squares to the data points

$$(x, y) = (-1, 7), \quad (1, 7), \quad (2, 21).$$

(a) Write the three equations $X\theta = y$, identifying X and y.

(b) Solve the normal equations for $\hat{\theta} = (\theta_0, \theta_1)^T$.

(c) Find the fitted values $\hat{y} = X\hat{\theta}$ and the residual vector $e = y - \hat{y}$.

(d) Verify that e is orthogonal to both columns of X.

Solution:

(a) The three equations are

$$\theta_0 - \theta_1 = 7, \qquad \theta_0 + \theta_1 = 7, \qquad \theta_0 + 2\theta_1 = 21,$$

so

$$X = \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad y = \begin{pmatrix} 7 \\ 7 \\ 21 \end{pmatrix}.$$

(b)

$$X^T X = \begin{pmatrix} 3 & 2 \\ 2 & 6 \end{pmatrix}, \qquad X^T y = \begin{pmatrix} 35 \\ 42 \end{pmatrix}.$$

Multiplying the first equation by 3 and subtracting the second gives $7\theta_0 = 63$, so $\theta_0 = 9$, and then $\theta_1 = 4$. The best line is $\hat{y} = 9 + 4x$.

(c) $\hat{y} = (5, 13, 17)^T$ and $e = (2, -6, 4)^T$, so $\|e\|_2^2 = 4 + 36 + 16 = 56$.

(d) Column 1: $2 - 6 + 4 = 0$. Column 2: $(-1)(2) + (1)(-6) + (2)(4) = 0$.

Problem 5

Find the best least-squares line $\hat{y} = \theta_0 + \theta_1 x$ for the data

$$y = (4, 2, -1, 0, 0)^T \quad \text{at} \quad x = -2, -1, 0, 1, 2.$$

(a) Compute $X^T X$ and $X^T y$, and solve for $\hat{\theta}$.

(b) Find the fitted values, the residual vector e, and $\|e\|_2^2$.

(c) Explain why $X^T X$ is diagonal for this data set, and what this means for solving for θ_0 and θ_1.

Solution:

(a)

$$X^T X = \begin{pmatrix} 5 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 10 \end{pmatrix}, \qquad X^T y = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = \begin{pmatrix} 5 \\ -10 \end{pmatrix}.$$

So $\theta_0 = 1$ and $\theta_1 = -1$, and the best line is $\hat{y} = 1 - x$.

(b) $\hat{y} = (3, 2, 1, 0, -1)^T$, $e = (1, 0, -2, 0, 1)^T$, and $\|e\|_2^2 = 6$.

(c) The inputs sum to zero, so the column of ones is orthogonal to the column of x values and the off-diagonal entries of $X^T X$ vanish. The two normal equations decouple and can be solved separately:

$$\theta_0 = \bar{y}, \qquad \theta_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2}.$$

Problem 6

(a)

Find the degree-2 polynomial $\hat{y} = \theta_0 + \theta_1 x + \theta_2 x^2$ that passes through the points $(x, y) = (0, 6),\ (1, 0),\ (2, 0)$.

(b) What are the fitted values and the residuals here? Why are the normal equations unnecessary?

(c) A fourth data point (x_4, y_4) is added that does not lie on this parabola. Will the new least-squares parabola keep zero residual at the first three points and put all the error at the fourth? Explain.

Solution:

(a) The equations are

$$\theta_0 = 6, \qquad \theta_0 + \theta_1 + \theta_2 = 0, \qquad \theta_0 + 2\theta_1 + 4\theta_2 = 0.$$

Substituting $\theta_0 = 6$ gives $\theta_1 + \theta_2 = -6$ and $\theta_1 + 2\theta_2 = -3$, so $\theta_2 = 3$ and $\theta_1 = -9$. The polynomial is $\hat{y} = 6 - 9x + 3x^2$.

(b) With three points and three parameters, X is a square, invertible matrix, so $X\theta = y$ is solved exactly: $\hat{y} = y$ and $e = 0$. No least-squares compromise is needed.

(c) No. The least-squares residual vector must be orthogonal to the first column of X, which is $(1, 1, 1, 1)^T$, so the four residuals must sum to zero. Least squares therefore spreads the error across all four points rather than putting it all at the new one.

Problem 7

Fit the degree-2 polynomial $\hat{y} = \theta_0 + \theta_1 x + \theta_2 x^2$ by least squares to the data

$$y = (0, 0, 1, 0, 0)^T \quad \text{at} \quad x = -2, -1, 0, 1, 2.$$

(a) Write the five equations $X\theta = y$ for the polynomial to pass through all five points. Explain why there is no exact solution.

(b) Compute X^TX. Before solving, predict the value of θ_1 and justify your prediction. *Hint:* notice what the symmetry of the inputs does to the power sums in X^TX.

(c) Solve the normal equations $X^TX\hat{\theta} = X^Ty$.

(d) Find the residual vector $e = y - X\hat{\theta}$ and $\|e\|_2^2$. Verify that e is orthogonal to every column of X.

Solution:

(a) Each row of X is $(1, x_i, x_i^2)$:

$$X = \begin{pmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}, \qquad y = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

There are five equations in three unknowns, and y is not in the column space of X: a polynomial that equals 1 at $x = 0$ and 0 at $x = \pm 1, \pm 2$ would need four roots, which a nonzero quadratic cannot have.

(b) The power sums are $\sum x_i^0 = 5$, $\sum x_i = 0$, $\sum x_i^2 = 10$, $\sum x_i^3 = 0$, $\sum x_i^4 = 34$, so

$$X^TX = \begin{pmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{pmatrix}.$$

Column 2 of X is orthogonal to columns 1 and 3, so the equation for θ_1 decouples as $10\theta_1 = \sum_i x_i y_i = 0$, giving $\theta_1 = 0$. This agrees with intuition: the data are symmetric about $x = 0$, so the best parabola should be too.

(c) $X^Ty = (1, 0, 0)^T$, since only the observation at $x = 0$ is nonzero. The remaining equations are

$$5\theta_0 + 10\theta_2 = 1, \qquad 10\theta_0 + 34\theta_2 = 0.$$

The second gives $\theta_0 = -3.4\,\theta_2$; substituting into the first gives $-7\theta_2 = 1$. Therefore

$$\theta_2 = -\frac{1}{7}, \qquad \theta_0 = \frac{17}{35}, \qquad \theta_1 = 0,$$

and the best parabola is $\hat{y} = \dfrac{17}{35} - \dfrac{1}{7}x^2$.

(d) The fitted values are $\hat{y}(0) = \frac{17}{35}$, $\hat{y}(\pm 1) = \frac{12}{35}$, $\hat{y}(\pm 2) = -\frac{3}{35}$, so

$$e = \frac{1}{35}(3,\ -12,\ 18,\ -12,\ 3)^T.$$

Orthogonality checks:

$$\sum_i e_i = 0, \qquad \sum_i x_i e_i = 0 \text{ (by symmetry)}, \qquad \sum_i x_i^2 e_i = \tfrac{1}{35}(12 - 12 + 0 - 12 + 12) = 0.$$

Therefore

$$\|e\|_2^2 = \frac{9 + 144 + 324 + 144 + 9}{1225} = \frac{630}{1225} = \frac{18}{35} \approx 0.51.$$

Problem 8

A data set of $n = 100$ observations has a single input feature X and a numerical output Y. Two models are fit by least squares: a linear model $Y = \beta_0 + \beta_1 X + \varepsilon$, and a cubic model $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$. For each model, let e_{train} be the vector of residuals $y_i - \hat{y}_i$ on the training data and e_{test} the vector of residuals on a separate test set.

(a) Suppose the true relationship between X and Y is linear. Compare the training error $\|e_{\text{train}}\|_2^2$ of the two models. Would you expect one to be lower, would you expect them to be equal, or is there not enough information to tell? Justify your answer.

(b) Answer (a) for the test error $\|e_{\text{test}}\|_2^2$ instead of $\|e_{\text{train}}\|_2^2$.

(c) Suppose the true relationship is not linear, but you do not know how far from linear it is. Compare $\|e_{\text{train}}\|_2^2$ for the two models, with justification.

(d) Answer (c) for $\|e_{\text{test}}\|_2^2$ instead of $\|e_{\text{train}}\|_2^2$.

Solution:

(a) The cubic model's $\|e_{\text{train}}\|_2^2$ will be lower than or equal to the linear model's. The linear model is the special case $\beta_2 = \beta_3 = 0$ of the cubic, so least squares over the larger family can never fit the training data worse. In practice it is strictly lower, because the extra terms fit some of the noise.

(b) The linear model is expected to have the lower $\|e_{\text{test}}\|_2^2$. The extra cubic terms reduce no bias (the truth is linear) but add variance by fitting noise, which hurts on new data. This is an expectation, not a guarantee for every sample.

(c) The cubic model's $\|e_{\text{train}}\|_2^2$ will again be lower than or equal to the linear model's, by the same nesting argument as in (a). This holds regardless of the true relationship.

(d) There is not enough information to tell. If the truth is far from linear, the cubic model's lower bias will likely give it the lower $\|e_{\text{test}}\|_2^2$. If the truth is only slightly nonlinear, the linear model's lower variance may win. In practice the choice is made using validation data.